

Ayush Patel

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EDUCATION

University of Michigan

Graduation: May 2026

BSE in Computer Science

Ann Arbor, MI

- **Relevant Coursework:** Discrete Mathematics, Data Structures & Algorithms, Computer Organization, Web Systems, Data-Driven Web Applications, Intro to AI, Foundations of Computer Science, Web Design, Info Retrieval & Web Search
- **Clubs & Organizations:** Michigan Data Science Team, Google Developer Group, Michigan Club Swim

EXPERIENCE

RIIS LLC

May 2025 – August 2025

Software Engineering Intern

Troy, MI

- Engineered a Python automation pipeline using OpenAI GPT-4o API, ElevenLabs TTS API, and FFmpeg to transcribe and summarize meetup recordings, generate AI voiceovers, and programmatically overlay narration onto slide decks; increased average viewer retention by 18%
- Developed an automated flight agent using the Amadeus Flight Offers Search API (v3), retrieving and ranking 200+ real-time global flight offers per query and automatically executing bookings based on user-defined constraints
- Implemented an automated workflow using n8n that processed sales, recruiter, and insurance-related data, applied business rules, and generated a finalized summary spreadsheet to streamline commission tracking and reporting

Miracle Software Systems

May 2024 – August 2024

Software Engineering Intern

Novi, MI

- Collaborated with a team of three interns to build and deploy 4 full-stack applications using Python, Streamlit, and GCP, supporting the full software lifecycle across API integration, real-time data processing, and conversational search
- Developed a BigQuery-powered GenAI chatbot enabling managers to look up employee information in natural language across 2,500+ records, using prompt engineering to improve query interpretation accuracy by 60%

BEET

June 2023 – August 2023

Software Engineering Intern

Southfield, MI

- Evaluated the viability of Apache Superset by conducting a thorough analysis of key metrics, laying the groundwork for a seamless transition, and providing strategic insights to optimize implementation and drive business impact
- Collected and analyzed datasets from 1,000+ factory machines using JavaScript and Apache Superset, transforming raw production data into actionable insights that improved process efficiency by 12%

PROJECTS

StudyBuddy | React, TypeScript, Node.js, Express, REST API, PostgreSQL, Cursor

- Built and deployed a full-stack study partner matching platform with a React frontend, Node.js/Express backend, and PostgreSQL database, using Cursor to accelerate feature development, debugging, and iteration
- Implemented a matching algorithm that calculates compatibility scores from shared courses, schedule overlap, study habits, campus frequency, and meeting preferences to rank recommended study partners
- Designed REST API endpoints and relational database schemas for authentication, user profiles, enrollments, study requests, direct messages, and course-based discussion features

Memento | Python, JavaScript, React, HTML, Flask, SQL, REST API, AWS

- Developed and deployed a full-stack social media web application with server-side dynamic pages using Flask, SQLite, and AWS, enabling seamless user authentication, content management, and real-time interactions
- Designed and integrated RESTful APIs to support client-side dynamic pages with React and Fetch API, implementing features such as infinite scrolling, interactive likes, and real-time comment updates

Youtube Explicit Content Filter | React, Firebase, Vertex AI

- Built a Chrome extension that filters explicit YouTube videos in real time by analyzing video captions and blocking content that violates family-friendly rules
- Implemented a caption-matching pipeline using a terms database + Vertex AI text embeddings for partial-match detection (94% accuracy) and added a user report/feedback flow to improve coverage over time

SKILLS

Languages & Concepts: Python, C++, JavaScript/TypeScript, SQL, HTML/CSS, Java, R, Object-Oriented Programming

AI Tools: OpenAI API, Claude, GCP Vertex AI, GitHub Copilot, LangChain, Cursor IDE, Prompt Engineering

Frameworks/Libraries: React, Node.js, Express, Flask, Django, Tailwind CSS, Pandas, Matplotlib

Cloud/Tools: GCP, AWS, BigQuery, PostgreSQL, Firebase, Docker, Postman, GitHub, VS Code

CERTIFICATES & TRAINING

Certificates: Google Artificial Intelligence Professional Certificate

Training & Badges: Salesforce Agentblazer Champion (*In Progress*); Anthropic Academy – Model Context Protocol (MCP), Subagents; Google Cloud Skills – Generative AI/LLMs; DeepLearning.AI - LangChain for LLM Application Development